3.6 Input and output devices

You now know how a computer processes data, but how does it receive that data and how do we see the result of the processing? This is possible through the use of input and **output devices**.

An **input device** is any device that allows data to be entered into a computer system. This data could take many forms including text, images and sound.

Common examples of input devices are:

- barcode scanner
- digital camera
- keyboard
- microphone
- optical mouse
- QR code scanner
- scanner (2D and 3D)
- touch screen (resistive, capacitive and infra-red).

KEY WORDS

output device: a device that can be used to obtain the results of data that has been processed.

input device: a device that allows data to be entered into a computer system.

Question

9 Copy the table below that includes the examples of the input devices. For each input device, describe the kind of data that it allows to be input and give an example of when it is used. The first one has been done for you.

Input device	Description of data input	Example of use
Barcode scanner	*It scans a barcode so that the data stored in the barcode can be obtained.*	*It is used in a supermarket to get the price of a product and as part of a stock control system.*
Digital camera		
Keyboard		
Microphone		
Optical mouse		
QR code scanner		
Scanner (2D and 3D)		
Touch screen (resistive, capacitive and infra-red)		

An output device is any device that allows the result of the data processing to be seen or heard.

Common examples of output devices are:

- actuator
- digital light processing (DLP) projector

- inkjet printer
- laser printer
- light emitting diode (LED) screen
- liquid crystal display (LCD) projector
- liquid crystal display (LCD) screen
- speaker
- 3D printer.

Question

10 Copy the table below that includes the examples of the output devices. For each output device, describe the kind of data that it allows to be output and give an example of when it is used. The first one has been done for you.

Output device	Description of data output	Example of use
Actuator	*It is a component that outputs an action, often a type of movement, that causes another device to operate.*	*Can be used in an automated system to move or turn on/off another device, e.g. a light.*
Digital light processing (DLP) projector		
Inkjet printer		
Laser printer		
Light emitting diode (LED) screen		
Liquid crystal display (LCD) projector		
Liquid crystal display (LCD) screen		
Speaker		
3D printer		

Many computer systems that you encounter in your daily life are automated systems. You can learn more about what an automated system is and how they work in Chapter 6.

A type of input device called a **sensor** is used in an **automated system**. The sensor is used to capture different types of data to be processed by the system. The sensor captures analogue data from its immediate surrounding environment. This data can then be processed and analysed by a computer system. Sensors are normally set to capture data at given time intervals. This could be as constant as every single second or even fraction of a second. It could also be set to be every minute or at particular times of the day.

Table 3.1 shows the different types of sensor that you need to know, a description of the type of data that they capture and an example of a how they might be used.

KEY WORDS

sensor: a type of input device that is used to capture data from its immediate environment.

automated system: a system that is designed to operate without the need for any human intervention.

Type of sensor	Type of data captured	Example of its use
Acoustic	This type of sensors measures the level of sound in an environment.	These sensors are used in many applications that involve sound. An acoustic sensor can be used in a security system. It can be placed near a window and constantly measure the level of sound. If it captures a reading that shows a sudden increase in sound, this could mean that the window has been broken and the building may be at risk.
Accelerometer	This type of sensor measures acceleration forces. These may be static forces, such as the continual force of gravity. They can also be dynamic forces, such as those created by movement and vibrations.	These sensors are used in a wide variety of devices. Mobile phones use an accelerometer to know which way up it is faced, to automatically turn the screen on and off. They can be used to monitor for earthquakes, as they can capture the initial vibrations created. They can also be used in cars to sense when a crash has occurred, so that airbags can be inflated.
Flow	This type of sensor measures the amount of liquid, gas or steam that is flowing through or around a certain environment.	These sensors are often used in factories and sites such as nuclear power plants. They make sure that the liquid, gas or steam flows at a constant temperature through an environment, such as a pipe. This makes sure that the pipes don't rupture and break due to too much flowing through them.
Gas	This type of sensor measures the presence and concentration of a gas within the immediate atmosphere.	These sensors can be used in people's homes. They can be set to measure a certain gas, such as carbon monoxide. They constantly capture the data in the immediate atmosphere to see if too much carbon monoxide is present, which could endanger the health of anyone living in the home.
Humidity	This type of sensor measures the level of moisture in the immediate atmosphere.	These sensors can be used in farming and agriculture to make sure that the air in areas such as greenhouses has the correct level of moisture to provide the best growing conditions for the fruits and vegetables. They can also be used in places such as art galleries, to make sure that the humidity level is constant. Too much or too little can ruin paintings.
Infra-red	This type of sensor measures infra-red radiation. This type of radiation can be emitted in different amounts by both objects and people.	These sensors can also be used in security systems. This can be done in two different ways. The device containing the sensor can emit infra-red radiation and when this bounces back to the device, the readings can show from the distance it has travelled whether an intruder is present. It can also operate by capturing the infra-red radiation emitted by the intruder.
Level	This type of sensor measures whether a substance, such as a liquid, is at a certain level or amount.	These sensors can be used in a car to make sure that essential liquids, such as oil and fuel, do not get too low.
Light	This type of sensor measures the ambient light in a certain environment. It can also measure the presence of a direct light, such as a laser beam.	These sensors can be used in automatic lighting systems. Streetlights can be fitted with a light sensor that will allow the light to turn on in the evening, when it becomes darker.

(continued)

Type of sensor	Type of data captured	Example of its use
Magnetic field	This type of sensor measures the presence of magnetic field that may be emitted by an object.	These sensors can be used to count how many cars pass through a certain area, for example into a car park. The car will disrupt the Earth's naturally magnetic field as it passes over the sensor and the data can be captured by the sensor.
Moisture	This type of sensor measures the amount of water that is present in a substance, such as soil.	These sensors can also be used by farming and agriculture, to make sure that the fruits and vegetables have the best level of water in the soil to help them grow.
pH	This type of sensor measures the pH level of a substance.	These sensors can be used by environmental agencies to make sure that local lakes and river are not being polluted.
Pressure	This type of sensor measures the force of pressure that is applied to the sensor or device. This could be the pressure created by a solid object, or it could be created by liquid or gas.	These sensors could also be used in a security system. They can be placed at the base or sides of an opening, such as a window or a door. When that window or door is opened, the pressure will decrease, and the system will recognise that an intruder has entered.
Proximity	This type of sensor measures how close an object is in comparison to the sensor. It does this by emitting electromagnetic radiation or an electromagnetic field and measures the radiation as it returns to see if there are any changes.	These types of sensors can be used in robots in manufacturing. They allow the robots to measure how close they are to different objects, when moving around a factory.
Temperature	This type of sensor measures the temperature of an object or substance by either directly touching it or capturing data from the surrounding environment.	These types of sensors are used in air conditioning systems. They allow the temperature of a room to be kept at a certain level.

Table 3.1: Different types of sensor



Figure 3.6: An accelerometer can sense a crash so that airbags inflate

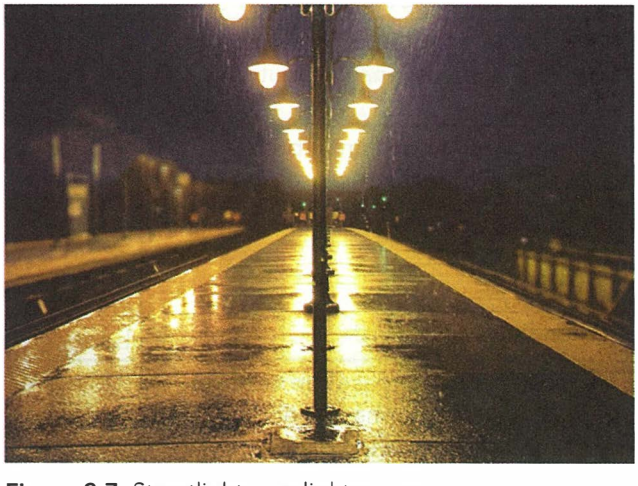


Figure 3.7: Streetlights use light sensors

Questions

11 Is a sensor an example of an input device or an output device?

12 Can you name **five** different types of sensor?

13 Can you think of one application for each of the following sensors?

- a Light
- b Magnetic field
- c Temperature

14 Can you think of **two** ways that the use of sensors is of benefit to humans?

15 How could an accelerometer be used to protect the hard drive on a computer?



Figure 3.8: Art galleries use temperature and humidity sensors

3.7 Data storage

You need to store data in a computer system on a constant basis. This might be temporarily, whilst it is being processed, or on a more permanent basis for future use. For these purposes, there are two main categories of storage that exist in a computer, they are **primary storage** and **secondary storage**.

Primary storage is a category of storage that is directly accessed by the CPU. This means that the CPU has a direct connection to it and that data is sent between the CPU and the primary storage using this connection. There are two main types of primary storage, random access memory (RAM) and **read only memory (ROM)**. There are several characteristics of each type of primary storage, these include those shown in Table 3.2:

Characteristics of RAM	Characteristics of ROM
It is volatile storage. This mean that it requires power to store data. If the power is turned off, the contents of RAM are lost. Volatile storage is sometimes referred to as temporary storage.	It is non-volatile. This means that if power is removed, the contents of the ROM are retained. Non-volatile storage is sometimes referred to as permanent storage.
It stores data and programs that are currently in use by the computer.	It stores programs that initially boot the computer, such as the bootstrap and the BIOS.
Its contents are constantly being changed as data is processed by the CPU.	The contents of ROM are fixed and do not normally change. It is possible to reprogram it, but this is a complex process and only normally attempted in circumstance when a computer system is failing to boot correctly.
It is possible to increase the amount of RAM a computer has by adding further RAM components	It is not easy to alter the ROM available in a computer system.

Table 3.2: Characteristics of primary storage

KEY WORDS

primary storage: a type of storage that is directly accessed by the CPU.

secondary storage: a type of storage that is designed to store data permanently, until it is deleted by the user.

ROM (read only memory): a type of primary storage that stores the start up instruction for the computer.